EDA with Netflix

Data Overview

Netflix is a leading global streaming platform that offers a vast library of movies, TV shows, and original content. This dataset provides insights into the platform's content offerings, including titles, genres, publication details, and ratings. It enables analysis of content trends, popularity, and audience preferences across different regions and time periods.

Importing Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
%matplotlib inline
```

Data Load

```
netflix = pd.read_csv('netflix_titles.csv')
```

Get the Top 10 rows

```
netflix.head(10)
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmmaker Kirsten Johnson stumbles upon moments she never imagined...
1	s2	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nathalie Emmanuel, Nicolas Duvauchelle, Laurent Lucas, Clémence Poésy	NaN	September 24, 2021	2021	TV-MA	1 Season	International TV Shows, TV Action & Adventure, TV Dramas, TV Mysteries	To protect his family from a powerful drug lord, a skilled thief and his crew must take down a ruthless...
2	s3	TV Show	Midnight Mass	Mike Flanagan	Kate Siegel, Zach Gilford, Hamish Linklater, Annabeth Gish, Alex Essoe, Rahul Kohli, Igal Naor, Samantha Sloyan, Gretchen Cox, Henry Thomas	NaN	September 24, 2021	2021	TV-MA	1 Season	TV Dramas, TV Horror, TV Mysteries	The Muslim community in a remote island village is thrilled when a young man returns home after a...
3	s4	Movie	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha	Vanessa Hudgens, Kimiko Glenn, James Marsden, Sofia Carson, Liza Koshy, Ken Jeong, Elizabeth Perkins	NaN	September 24, 2021	2021	PG	90 min	Children & Family Movies, Comedies	Some years after the events of the first film, Princess Twilight Sparkle and her friends are living in...
4	s5	Movie	Sankofa	Gaston Kabore	NaN	NaN	September 24, 2021	1993	TV-MA	125 min	Dramas, Independent Movies, International Movies	Mixing fiction and dreamlike sequences, this film follows a model who is photographed by a famous...
5	s6	TV Show	The Great British Baking Show	Andy Devonshire	Mel Giedroyc, Sue Perkins, Mary Berry, Paul Hollywood	United Kingdom	September 24, 2021	2021	TV-PG	9 Seasons, 53 Episodes	British TV Shows, Reality TV	Bake-off fever sweeps the nation as 12 new contestants vie for the title of Britain's best home baker...
6	s7	Movie	The Starling	Theodore Melfi	Melissa McCarthy, Chris O'Dowd, Kevin Kline, Timothy Olyphant, Daveed Diggs, Skyler Gisondo, Loretta Devine, Laura Harrier, Rosalind Chao, Dave Bautista	United States	September 23, 2021	2021	PG-13	104 min	Comedies, Dramas	Lily and her husband struggle to deal with a devastating loss. Lily's attempts to find her way back to...
7	s8	Movie	Blindspotting	Carlos Lopez Estrada	Rafael Casal, Jasmine Cephas-Jones, Jaylin Fridvall, Ethan Ramos, Tiffany Mack, Utkarsh Ambudkar, Janelle Monáe, Jillian Bell, Brett Gelman, Scott Lawrence	United States	September 23, 2021	2018	R	95 min	Comedies, Dramas, Independent Movies	Collin (Daveed Diggs) has lived in Oakland his entire life. As he ticks off the days to his release from...
8	s9	Movie	Jagame Thandhiram	Karthik Subbaraj	Dhanush, James Cosmo, Jeffrey Dean Morgan, Alister, Joe Pantoliano, Kalaiyarasan, Nassar, Riya Vidyarthi, Sanchana Natarajan, Shanthi Krishna, Vivek Prasanna, Yogi Babu	India	September 23, 2021	2021	TV-MA	157 min	Dramas, International Movies, Music & Musicals	When a gangster faces a murder charge, one of his men is persuaded to take the rap in exchange for a...
9	s10	TV Show	Eaten by the Newt	NaN	Ashleigh Cummings, Thomas Cavanagh, Jason Dhanveer, Kiana Madeira, Steven Ogg, Scott Bakula	NaN	September 23, 2021	2021	TV-MA	1 Season	TV Comedies, TV Dramas	The survivors of a plane crash are forced into a deadly game of cat and mouse when they crash land in the...

```
netflix.tail(10)
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
8797	s8798	TV Show	Trinkets	NaN	Kiernan Shipka, Matthew Lawrence, Jessica Rothe, Nick Eversman, Ashlie Atkinson, Ryan Cartwright, Ciara Bravo, Elena Kampouris, Jordan Fisher, Anna Cathcart, Tom Ellis, Kate Siegel, Jackie Tohn, Val Emmich, Fortune Feimster	NaN	December 15, 2017	2017	TV-MA	2 Seasons	TV Comedies, TV Dramas, TV Romantic Comedies	After her best friend's death, a high school girl and her two friends discover her secret diary, which leads them on a quest to her Mexican hometown.
8798	s8799	Movie	Trinkets	NaN	NaN	NaN	December 15, 2017	2017	TV-MA	90 min	Comedies, Dramas, Independent Movies	After her best friend's death, a high school girl and her two friends discover her secret diary, which leads them on a quest to her Mexican hometown.
8799	s8800	Movie	Vendetta: Truth, Lies and The Mafia	Sam Vaughan	NaN	NaN	December 15, 2017	2017	TV-MA	60 min	Documentaries, International Movies	Generations after the Sicilian Mafia's bloody rampage, a young man who's lost his father to the cause...
8800	s8801	TV Show	Vendetta: Truth, Lies and The Mafia	NaN	NaN	NaN	December 15, 2017	2017	TV-MA	1 Season	Crime TV Shows, Docuseries, International TV Shows	Generations after the Sicilian Mafia's bloody rampage, a young man who's lost his father to the cause...
8801	s8802	Movie	Zinzana	Majid Al Ansari	Ali Suliman, Saleh Bakri, Yousef Sweidan, Amina Khalil, Fares Fares, Assaad Bouab, Antonio De La Torre, Raúl Arévalo	United Arab Emirates, Jordan, Syria, Lebanon, Spain, Morocco, Iraq	December 15, 2017	2015	TV-MA	96 min	Dramas, International Movies, Thrillers	Amidst the chaos and bloodshed of the Lebanese Civil War, a young man is recruited by a shadowy...
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey Jr., Anthony Edwards, Brian Cox, Elias Koteas, Joel Gretsch, Jack Nance, Donal Logue, William Sadler, Bob Stephenson, Chloë Sevigny, Russell Williams II, Charles Fleischer, Lee Norris	United States	November 15, 2017	2007	R	158 min	Cult Movies, Dramas, Thrillers	A dogged cartoonist, a jaded journalist, and a persistent detective try to track down a serial killer who...
8803	s8804	TV Show	Zoom	NaN	Jim Rash, Fred Armisen, Kate Micucci, Gregg Turkington, Credence Cleaver, Tyler Labine, Chevy Chase, Artemis Pebdenis, Cameron Ocasio, Ariana Greenblatt, Michael Cera, Tituss Burgess, Concetta Tomei, Ryan Cartwright, Ryan Lee, Kate Micucci, Gregg Turkington, Credence Cleaver, Tyler Labine, Chevy Chase, Artemis Pebdenis, Cameron Ocasio, Ariana Greenblatt, Michael Cera, Tituss Burgess, Concetta Tomei, Ryan Cartwright, Ryan Lee	NaN	November 15, 2017	2017	TV-PG	90 min	Children & Family Movies, Comedies	Frustrated by their parents' lack of interest, a young girl and her friends decide to take their...

Get information of the data

```
netflix.info()
```

```
<class type="dataset_info">
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 show_id       8807 non-null object
 type          8807 non-null object
 title         8807 non-null object
 director      6173 non-null object
 cast          7982 non-null object
 country       7976 non-null object
 date_added    8797 non-null object
 release_year  8807 non-null int64
 rating        8803 non-null object
 duration      8804 non-null object
 listed_in     8807 non-null object
 description   8807 non-null object
dtype: int64, object: 11
memory usage: 1.1+ MB
</class type="dataset_info">
```

First Null Values

```
netflix.isnull().sum()
```

```
show_id       0
type          0
title         0
director      2634
cast          831
country       829
date_added    10
release_year  0
rating        4
duration      3
listed_in     0
description   0
dtype: int64
```

Check sum and check Unique

```
netflix.nunique()
```

```
show_id       8807
type          2
title         8807
director      4528
cast          7692
country       748
date_added    1767
release_year  73
rating        14
duration      220
listed_in     514
description   8775
dtype: int64
```

Drop null

```
netflix.dropna(inplace=True)
```

Check null values in release year column

```
netflix['release_year'].isnull()
```

```
0       False
1       False
2       False
3       False
4       False
5       False
6       False
7       False
8       False
9       False
8797    False
8798    False
8799    False
8800    False
8801    False
8802    False
8803    False
Name: release_year, dtype: bool
```

Box Plot

Box plot for release_year and rating

```
sns.boxplot(netflix['release_year'])
```

```
sns.boxplot(netflix['rating'])
```

Count plot

Count plot of Rating column after handling Outliers

```
sns.countplot(netflix['rating'])
```


Count of Release Year column after handling Outliers

```
sns.countplot(netflix['release_year'])
```

Checking outliers in release_year

```
sns.boxplot(netflix['release_year'])
```

Boxplot of Release Year column after handling Outliers

```
sns.boxplot(netflix['release_year'])
```


Since release_rating_year column dropped so there was no need to change type from int to

```
netflix.dtypes
```

```
show_id       object
type          object
title         object
director      object
cast          object
country       object
date_added    object
release_year  int64
rating        object
duration      object
listed_in     object
description   object
dtype: object
```

1. Distribution of content

Distribution of content by type (movies or tv shows) over different genres.

```
netflix['listed_in'].value_counts()
```

```
International Movies                                        2624
Dramas                                                      1600
Comedies                                                    1210
Documentaries                                               1000
Dramas, International Movies                                 362
Comedies, Dramas                                             359
Comedies, International Movies                               358
Dramas, Independent Movies                                   334
Kids' TV                                                     312
Children & Family Movies                                     290
Comedies, Dramas, International Movies                       274
Comedies, Dramas, Independent Movies                         252
Kids' TV                                                     241
Independent Movies                                           231
International TV Shows, TV Dramas                            219
Thrillers                                                    195
International TV Shows, TV Comedies                          193
Crime TV Shows, International TV Shows, TV Dramas            131
Action & Adventure                                           128
Romantic TV Shows, TV Comedies, TV Dramas                    126
Crime TV Shows, Docuseries, International TV Shows           118
Docuseries                                                   105
Comedies, Horror Movies                                      103
Dramas, Independent Movies, Thrillers                        101
Comedies, Dramas, Independent Movies, Romantic Movies        100
Comedies, Dramas, Independent Movies, Thrillers               97
Comedies, Dramas, Independent Movies, Romantic Movies         96
Comedies, Dramas, Independent Movies, Thrillers               95
Comedies, Dramas, Independent Movies, Romantic Movies         94
Comedies, Dramas, Independent Movies, Thrillers               93
Comedies, Dramas, Independent Movies, Romantic Movies         92
Comedies, Dramas, Independent Movies, Thrillers               91
Comedies, Dramas, Independent Movies, Romantic Movies         90
Comedies, Dramas, Independent Movies, Thrillers               89
Comedies, Dramas, Independent Movies, Romantic Movies         88
Comedies, Dramas, Independent Movies, Thrillers               87
Comedies, Dramas, Independent Movies, Romantic Movies         86
Comedies, Dramas, Independent Movies, Thrillers               85
Comedies, Dramas, Independent Movies, Romantic Movies         84
Comedies, Dramas, Independent Movies, Thrillers               83
Comedies, Dramas, Independent Movies, Romantic Movies         82
Comedies, Dramas, Independent Movies, Thrillers               81
Comedies, Dramas, Independent Movies, Romantic Movies         80
Comedies, Dramas, Independent Movies, Thrillers               79
Comedies, Dramas, Independent Movies, Romantic Movies         78
Comedies, Dramas, Independent Movies, Thrillers               77
Comedies, Dramas, Independent Movies, Romantic Movies         76
Comedies, Dramas, Independent Movies, Thrillers               75
Comedies, Dramas, Independent Movies, Romantic Movies         74
Comedies, Dramas, Independent Movies, Thrillers               73
Comedies, Dramas, Independent Movies, Romantic Movies         72
Comedies, Dramas, Independent Movies, Thrillers               71
Comedies, Dramas, Independent Movies, Romantic Movies         70
Comedies, Dramas, Independent Movies, Thrillers               69
Comedies, Dramas, Independent Movies, Romantic Movies         68
Comedies, Dramas, Independent Movies, Thrillers               67
Comedies, Dramas, Independent Movies, Romantic Movies         66
Comedies, Dramas, Independent Movies, Thrillers               65
Comedies, Dramas, Independent Movies, Romantic Movies         64
Comedies, Dramas, Independent Movies, Thrillers               63
Comedies, Dramas, Independent Movies, Romantic Movies         62
Comedies, Dramas, Independent Movies, Thrillers               61
Comedies, Dramas, Independent Movies, Romantic Movies         60
Comedies, Dramas, Independent Movies, Thrillers               59
Comedies, Dramas, Independent Movies, Romantic Movies         58
Comedies, Dramas, Independent Movies, Thrillers               57
Comedies, Dramas, Independent Movies, Romantic Movies         56
Comedies, Dramas, Independent Movies, Thrillers               55
Comedies, Dramas, Independent Movies, Romantic Movies         54
Comedies, Dramas, Independent Movies, Thrillers               53
Comedies, Dramas, Independent Movies, Romantic Movies         52
Comedies, Dramas, Independent Movies, Thrillers               51
Comedies, Dramas, Independent Movies, Romantic Movies         50
Comedies, Dramas, Independent Movies, Thrillers               49
Comedies, Dramas, Independent Movies, Romantic Movies         48
Comedies, Dramas, Independent Movies, Thrillers               47
Comedies, Dramas, Independent Movies, Romantic Movies         46
Comedies, Dramas, Independent Movies, Thrillers               45
Comedies, Dramas, Independent Movies, Romantic Movies         44
Comedies, Dramas, Independent Movies, Thrillers               43
Comedies, Dramas, Independent Movies, Romantic Movies         42
Comedies, Dramas, Independent Movies, Thrillers               41
Comedies, Dramas, Independent Movies, Romantic Movies         40
Comedies, Dramas, Independent Movies, Thrillers               39
Comedies, Dramas, Independent Movies, Romantic Movies         38
Comedies, Dramas, Independent Movies, Thrillers               37
Comedies, Dramas, Independent Movies, Romantic Movies         36
Comedies, Dramas, Independent Movies, Thrillers               35
Comedies, Dramas, Independent Movies, Romantic Movies         34
Comedies, Dramas, Independent Movies, Thrillers               33
Comedies, Dramas, Independent Movies, Romantic Movies         32
Comedies, Dramas, Independent Movies, Thrillers               31
Comedies, Dramas, Independent Movies, Romantic Movies         30
Comedies, Dramas, Independent Movies, Thrillers               29
Comedies, Dramas, Independent Movies, Romantic Movies         28
Comedies, Dramas, Independent Movies, Thrillers               27
Comedies, Dramas, Independent Movies, Romantic Movies         26
Comedies, Dramas, Independent Movies, Thrillers               25
Comedies, Dramas, Independent Movies, Romantic Movies         24
Comedies, Dramas, Independent Movies, Thrillers               23
Comedies, Dramas, Independent Movies, Romantic Movies         22
Comedies, Dramas, Independent Movies, Thrillers               21
Comedies, Dramas, Independent Movies, Romantic Movies         20
Comedies, Dramas, Independent Movies, Thrillers               19
Comedies, Dramas, Independent Movies, Romantic Movies         18
Comedies, Dramas, Independent Movies, Thrillers               17
Comedies, Dramas, Independent Movies, Romantic Movies         16
Comedies, Dramas, Independent Movies, Thrillers               15
Comedies, Dramas, Independent Movies, Romantic Movies         14
Comedies, Dramas, Independent Movies, Thrillers               13
Comedies, Dramas, Independent Movies, Romantic Movies         12
Comedies, Dramas, Independent Movies, Thrillers               11
Comedies, Dramas, Independent Movies, Romantic Movies         10
Comedies, Dramas, Independent Movies, Thrillers                9
Comedies, Dramas, Independent Movies, Romantic Movies          8
Comedies, Dramas, Independent Movies, Thrillers                7
Comedies, Dramas, Independent Movies, Romantic Movies          6
Comedies, Dramas, Independent Movies, Thrillers                5
Comedies, Dramas, Independent Movies, Romantic Movies          4
Comedies, Dramas, Independent Movies, Thrillers                3
Comedies, Dramas, Independent Movies, Romantic Movies          2
Comedies, Dramas, Independent Movies, Thrillers                1
Comedies, Dramas, Independent Movies, Romantic Movies          1
Name: listed_in, dtype: int64
```

Top 10 Genres Distribution

```
netflix['listed_in'].value_counts().head(10).plot(kind='pie')
```

```
netflix['listed_in'].value_counts()
```

```
International Movies                                        2624
Dramas                                                      1600
Comedies                                                    1210
Documentaries                                               1000
Dramas, International Movies                                 362
Comedies, Dramas                                             359
Comedies, International Movies                               358
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Kids' TV                                                     312
Children & Family Movies                                     290
Comedies, Dramas, International Movies                       274
Comedies, Dramas, Independent Movies                         252
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Independent Movies                                           231
International TV Shows, TV Dramas                            219
Thrillers                                                    195
International TV Shows, TV Comedies                          193
Crime TV Shows, International TV Shows, TV Dramas            131
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Romantic TV Shows, TV Comedies, TV Dramas                    126
Crime TV Shows, Docuseries, International TV Shows           118
Docuseries                                                   105
Comedies, Horror Movies                                      103
Dramas, Independent Movies, Thrillers                        101
Comedies, Dramas, Independent Movies, Romantic Movies        100
Comedies, Dramas, Independent Movies, Thrillers               97
Comedies, Dramas, Independent Movies, Romantic Movies         96
Comedies, Dramas, Independent Movies, Thrillers               95
Comedies, Dramas, Independent Movies, Romantic Movies         94
Comedies, Dramas, Independent Movies, Thrillers               93
Comedies, Dramas, Independent Movies, Romantic Movies         92
Comedies, Dramas, Independent Movies, Thrillers               91
Comedies, Dramas, Independent Movies, Romantic Movies         90
Comedies, Dramas, Independent Movies, Thrillers               89
Comedies, Dramas, Independent Movies, Romantic Movies         88
Comedies, Dramas, Independent Movies, Thrillers               87
Comedies, Dramas, Independent Movies, Romantic Movies         86
Comedies, Dramas, Independent Movies, Thrillers               85
Comedies, Dramas, Independent Movies, Romantic Movies         84
Comedies, Dramas, Independent Movies, Thrillers               83
Comedies, Dramas, Independent Movies, Romantic Movies         82
Comedies, Dramas, Independent Movies, Thrillers               81
Comedies, Dramas, Independent Movies, Romantic Movies         80
Comedies, Dramas, Independent Movies, Thrillers               79
Comedies, Dramas, Independent Movies, Romantic Movies         78
Comedies, Dramas, Independent Movies, Thrillers               77
Comedies, Dramas, Independent Movies, Romantic Movies         76
Comedies, Dramas, Independent Movies, Thrillers               75
Comedies, Dramas, Independent Movies, Romantic Movies         74
Comedies, Dramas, Independent Movies, Thrillers               73
Comedies, Dramas, Independent Movies, Romantic Movies         72
Comedies, Dramas, Independent Movies, Thrillers               71
Comedies, Dramas, Independent Movies, Romantic Movies         70
Comedies, Dramas, Independent Movies, Thrillers               69
Comedies, Dramas, Independent Movies, Romantic Movies         68
Comedies, Dramas, Independent Movies, Thrillers               67
Comedies, Dramas, Independent Movies, Romantic Movies         66
Comedies, Dramas, Independent Movies, Thrillers               65
Comedies, Dramas, Independent Movies, Romantic Movies         64
Comedies, Dramas, Independent Movies, Thrillers               63
Comedies, Dramas, Independent Movies, Romantic Movies         62
Comedies, Dramas, Independent Movies, Thrillers               61
Comedies, Dramas, Independent Movies, Romantic Movies         60
Comedies, Dramas, Independent Movies, Thrillers               59
Comedies, Dramas, Independent Movies, Romantic Movies         58
Comedies, Dramas, Independent Movies, Thrillers               57
Comedies, Dramas, Independent Movies, Romantic Movies         56
Comedies, Dramas, Independent Movies, Thrillers               55
Comedies, Dramas, Independent Movies, Romantic Movies         54
Comedies, Dramas, Independent Movies, Thrillers               53
Comedies, Dramas, Independent Movies, Romantic Movies         52
Comedies, Dramas, Independent Movies, Thrillers               51
Comedies, Dramas, Independent Movies, Romantic Movies         50
Comedies, Dramas, Independent Movies, Thrillers               49
Comedies, Dramas, Independent Movies, Romantic Movies         48
Comedies, Dramas, Independent Movies, Thrillers               47
Comedies, Dramas, Independent Movies, Romantic Movies         46
Comedies, Dramas, Independent Movies, Thrillers               45
Comedies, Dramas, Independent Movies, Romantic Movies         44
Comedies, Dramas, Independent Movies, Thrillers               43
Comedies, Dramas, Independent Movies, Romantic Movies         42
Comedies, Dramas, Independent Movies, Thrillers               41
Comedies, Dramas, Independent Movies, Romantic Movies         40
Comedies, Dramas, Independent Movies, Thrillers               39
Comedies, Dramas, Independent Movies, Romantic Movies         38
Comedies, Dramas, Independent Movies, Thrillers               37
Comedies, Dramas, Independent Movies, Romantic Movies         36
Comedies, Dramas, Independent Movies, Thrillers               35
Comedies, Dramas, Independent Movies, Romantic Movies         34
Comedies, Dramas, Independent Movies, Thrillers               33
Comedies, Dramas, Independent Movies, Romantic Movies         32
Comedies, Dramas, Independent Movies, Thrillers               31
Comedies, Dramas, Independent Movies, Romantic Movies         30
Comedies, Dramas, Independent Movies, Thrillers               29
Comedies, Dramas, Independent Movies, Romantic Movies         28
Comedies, Dramas, Independent Movies, Thrillers               27
Comedies, Dramas, Independent Movies, Romantic Movies         26
Comedies, Dramas, Independent Movies, Thrillers               25
Comedies, Dramas, Independent Movies, Romantic Movies         24
Comedies, Dramas, Independent Movies, Thrillers               23
Comedies, Dramas, Independent Movies, Romantic Movies         22
Comedies, Dramas, Independent Movies, Thrillers               21
Comedies, Dramas, Independent Movies, Romantic Movies         20
Comedies, Dramas, Independent Movies, Thrillers               19
Comedies, Dramas, Independent Movies, Romantic Movies         18
Comedies, Dramas, Independent Movies, Thrillers               17
Comedies, Dramas, Independent Movies, Romantic Movies         16
Comedies, Dramas, Independent Movies, Thrillers               15
Comedies, Dramas, Independent Movies, Romantic Movies         14
Comedies, Dramas, Independent Movies, Thrillers               13
Comedies, Dramas, Independent Movies, Romantic Movies         12
Comedies, Dramas, Independent Movies, Thrillers               11
Comedies, Dramas, Independent Movies, Romantic Movies         10
Comedies, Dramas, Independent Movies, Thrillers                9
Comedies, Dramas, Independent Movies, Romantic Movies          8
Comedies, Dramas, Independent Movies, Thrillers                7
Comedies, Dramas, Independent Movies, Romantic Movies          6
Comedies, Dramas, Independent Movies, Thrillers                5
Comedies, Dramas, Independent Movies, Romantic Movies          4
Comedies, Dramas, Independent Movies, Thrillers                3
Comedies, Dramas, Independent Movies, Romantic Movies          2
Comedies, Dramas, Independent Movies, Thrillers                1
Comedies, Dramas, Independent Movies, Romantic Movies          1
Name: listed_in, dtype: int64
```

Distribution of Content Over Different Genres

```
netflix['listed_in'].value_counts().plot(kind='bar')
```



```
netflix['listed_in'].value_counts().head(10).plot(kind='bar')
```


Top 10 Genres Distribution

Distribution of content across release years

```
netflix['release_year'].value_counts()
```

```
2018    1147
2017    1032
2016    902
2019    859
2020    857
2015    827
2014    745
2013    592
2012    411
2011    333
2010    321
2009    291
2008    274
2007    265
2006    231
2005    191
2004    176
2003    146
2002    119
2001    110
2000    96
1999    80
1998    63
1997    47
1996    43
1995    38
1994    27
1993    24
1992    23
1991    21
1990    16
1989    13
1988    11
1987    9
1986    8
1985    7
1984    6
1983    6
1982    4
1981    3
1980    3
1979    2
1978    2
1977    1
1976    1
1975    1
1974    1
1973    1
1972    1
1971    1
1970    1
1969    1
1968    1
1966    1
Name: release_year, dtype: int64
```

Distribution of Content Across Release Years

```
netflix['release_year'].value_counts().plot(kind='bar')
```

Distribution of content rating

```
netflix['rating'].value_counts()
```

```
TV-MA    3207
TV-14    2160
TV-PG    863
R        799
PG-13    490
PG       287
TV-Y7    334
TV-Y     307
TV-G     220
G        41
NC-17    3
NR       80
TV-Y7-FV 5
UR       3
Name: rating, dtype: int64
```

Distribution of Content Ratings

```
netflix['rating'].value_counts().plot(kind='bar')
```

Content TV-MA has most number of titles.

Time Series Analysis

Content added to netflix platform over the years (number of titles added per year)

```
netflix['date_added'].value_counts()
```

```
January 1, 2020      110
November 1, 2019     89
March 1, 2018        75
December 31, 2019    74
October 1, 2018      71
September 1, 2018    70
August 1, 2018       68
July 1, 2018         65
June 1, 2018         62
April 1, 2018        60
May 1, 2018          59
February 1, 2018     56
January 1, 2018      53
December 1, 2017     43
November 30, 2018    43
October 31, 2018     42
September 30, 2018   41
August 30, 2018      40
July 30, 2018        39
June 30, 2018        37
April 30, 2018       36
May 30, 2018         35
February 28, 2018    34
January 30, 2018     33
December 30, 2017    32
November 29, 2018    31
October 29, 2018     30
September 29, 2018   29
August 29, 2018      28
July 29, 2018        27
June 29, 2018        26
April 29, 2018       25
May 29, 2018         24
February 27, 2018    23
January 29, 2018     22
December 29, 2017    21
November 28, 2018    20
October 28, 2018     19
September 28, 2018   18
August 28, 2018      17
July 28, 2018        16
June 28, 2018        15
April 28, 2018       14
May 28, 2018         13
February 26, 2018    12
January 28, 2018     11
December 28, 2017    10
November 27, 2018    9
October 27, 2018     8
September 27, 2018   7
August 27, 2018      6
July 27, 2018        5
June 27, 2018        4
April 27, 2018       3
May 27, 2018         2
February 25, 2018    1
January 27, 2018     1
Name: date_added, dtype: int64
```

Number of Movies and TV Shows Released Per Year

```
netflix['release_year'].value_counts().head(10).plot(kind='bar')
```

Content Added by Year of Year

```
netflix['date_added'].value_counts().head(10).plot(kind='bar')
```

Content Added by Year of Year Movie

```
netflix[netflix['type'].isin(['Movie'])]['date_added'].value_counts().head(10).plot(kind='bar')
```

Top TV Shows

```
netflix[netflix['type'].isin(['TV Show'])]['date_added'].value_counts().head(10).plot(kind='bar')
```

Content added to netflix platform over the years (number of titles added per year) Movie and TV Shows

```
netflix[netflix['type'].isin(['Movie'])]['release_year'].value_counts()
```

```
2017    767
2018    767
2016    765
2019    633
2020    562
2015    560
2014    517
2013    411
2012    299
2011    256
2010    215
2009    201
2008    152
2007    139
2006    128
2005    116
2004    99
2003    90
2002    70
2001    62
2000    51
1999    37
1998    33
1997    26
1996    20
1995    15
1994    13
1993    13
1992    7
1991    8
1990    8
1989    6
1988    5
1987    3
1986    2
1985    3
1984    1
1983    2
1982    1
1981    1
1980    1
1979    1
1978    1
1977    1
1976    1
1975    1
1974    1
1973    1
1972    1
1971    1
1970    1
1969    1
1968    1
1966    1
Name: release_year, dtype: int64
```

Number of Movies in TV Shows

```
netflix[netflix['type'].isin(['TV Show'])]['release_year'].value_counts()
```

```
netflix[netflix['type'].isin(['TV Show'])]['release_year'].value_counts()
```

```
2020    295
2019    226
2018    180
2017    165
2016    137
2015    67
2014    28
2013    18
2012    12
2011    7
2010    6
2009    4
2008    3
2007    2
2006    1
2005    1
2004    1
2003    1
2001    1
1999    1
1998    1
1997    1
1996    1
1995    1
1993    1
1991    1
1989    1
1988    1
1986    1
1985    1
1983    1
1980    1
1979    1
1976    1
1975    1
1974    1
1973    1
1972    1
1971    1
1970    1
1969    1
1968    1
Name: release_year, dtype: int64
```

Top Lists and Recommendations

Identify and provide top lists (movies or TV shows based on user ratings

```
netflix[netflix['type'].isin(['Movie'])]['rating'].value_counts()
```

```
TV-MA    1931
TV-14    1143
R        797
PG-13    490
PG       287
TV-PG    271
TV-Y7    139
TV-Y     131
TV-G     126
G        41
NR       75
NC-17    3
UR       3
Name: rating, dtype: int64
```

Distribution of TV Shows by Rating

```
netflix[netflix['type'].isin(['TV Show'])]['rating'].value_counts()
```

Distribution of TV Shows by Rating

```
netflix[netflix['type'].isin(['TV Show'])]['rating'].value_counts().plot(kind='bar')
```

Genre Trends

Analyze trends in the popularity of different genres over time

```
netflix['listed_in'].value_counts()
```

```
International Movies                                        2624
Dramas                                                      1600
Comedies                                                    1210
Documentaries                                               1000
Dramas, International Movies                                 362
Comedies, Dramas                                             359
Comedies, International Movies                               358
Dramas, Independent Movies                                   334
Kids' TV                                                     312
Children & Family Movies                                     290
Comedies, Dramas, International Movies                       274
Comedies, Dramas, Independent Movies                         252
Kids' TV                                                     241
Independent Movies                                           231
International TV Shows, TV Dramas                            219
Thrillers                                                    195
International TV Shows, TV Comedies                          193
Crime TV Shows, International TV Shows, TV Dramas            131
Action & Adventure                                           128
Romantic TV Shows, TV Comedies, TV Dramas                    126
Crime TV Shows, Docuseries, International TV Shows           118
Docuseries                                                   105
Comedies, Horror Movies                                      103
Dramas, Independent Movies, Thrillers                        101
Comedies, Dramas, Independent Movies, Romantic Movies        100
Comedies, Dramas, Independent Movies, Thrillers               97
Comedies, Dramas, Independent Movies, Romantic Movies         96
Comedies, Dramas, Independent Movies, Thrillers               95
Comedies, Dramas, Independent Movies, Romantic Movies         94
Comedies, Dramas, Independent Movies, Thrillers               93
Comedies, Dramas, Independent Movies, Romantic Movies         92
Comedies, Dramas, Independent Movies, Thrillers               91
Comedies, Dramas, Independent Movies, Romantic Movies         90
Comedies, Dramas, Independent Movies, Thrillers               89
Comedies, Dramas, Independent Movies, Romantic Movies         88
Comedies, Dramas, Independent Movies, Thrillers               87
Comedies, Dramas, Independent Movies, Romantic Movies         86
Comedies, Dramas, Independent Movies, Thrillers               85
Comedies, Dramas, Independent Movies, Romantic Movies         84
Comedies, Dramas, Independent Movies, Thrillers               83
Comedies, Dramas, Independent Movies, Romantic Movies         82
Comedies, Dramas, Independent Movies, Thrillers               81
Comedies, Dramas, Independent Movies, Romantic Movies         80
Comedies, Dramas, Independent Movies, Thrillers               79
Comedies, Dramas, Independent Movies, Romantic Movies         78
Comedies, Dramas, Independent Movies, Thrillers               77
Comedies, Dramas, Independent Movies, Romantic Movies         76
Comedies, Dramas, Independent Movies, Thrillers               75
Comedies, Dramas, Independent Movies, Romantic Movies         74
Comedies, Dramas, Independent Movies, Thrillers               73
Comedies, Dramas, Independent Movies,
```